

Electrical Machine- II (BT EE403A)



हरियाणा केंद्रीय विश्वविद्यालय
CENTRAL UNIVERSITY OF HARYANA
NAAC ACCREDITED 'A' GRADE UNIVERSITY

Unit-IV

Synchronous Machine

- ❖ Concept of Synchronous Reactance
- ❖ Approximation of the Saturated Synchronous Reactance
 - Open-circuit and Short-circuit Tests
 - Open-circuit and Short-circuit Characteristic
- ❖ Unsaturated Synchronous Impedance

By
Dr. Manish Kumar
Asth. Professor
Department of Electrical Engineering
SOE&T, CUH

Concept of Synchronous Reactance

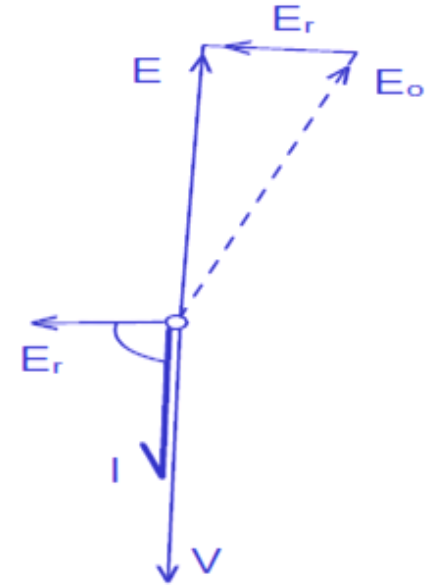
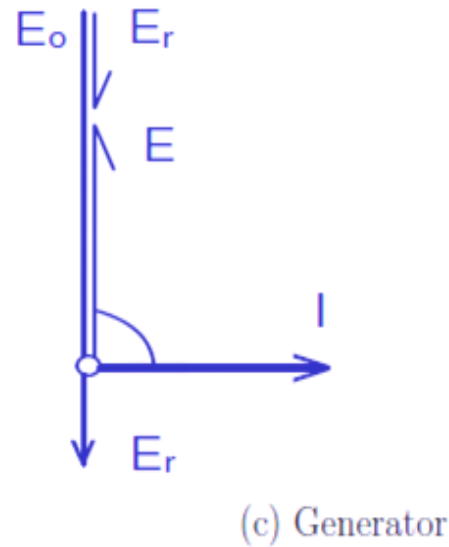
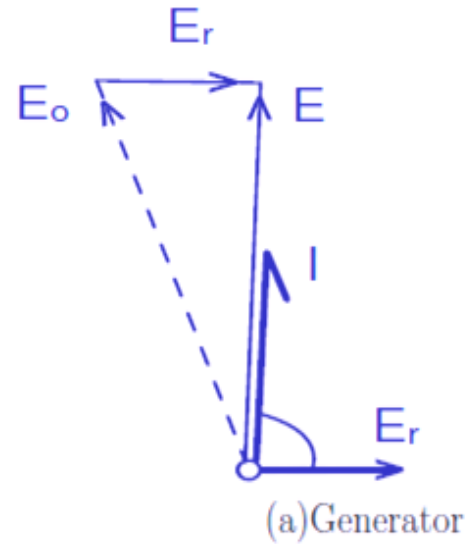
The operation of the synchronous machine can be reduced to comparatively simple expression by the convenient concept of synchronous reactance. The resultant linkage of flux with any phase of the armature of a synchronous machine is due, as has been seen, to the combined action of the field and armature currents. In brief the simplifying assumptions are...

1. The permeability of all parts of the magnetic circuit of the synchronous machine is constant - in other words the field and armature fluxes can be treated separately as proportional to their respective currents so that their effects can be superposed.
2. The air gap is uniform, so that the armature flux is not affected by its position relative to the poles - in other words we assume the rotor to be cylindrical.
3. The distribution of the field flux in the air gap is sinusoidal.
4. The armature winding is uniformly distributed and carries balanced sinusoidal currents:- in other words, the harmonics are neglected so that the armature flux is directly proportional to the fundamental component of the armature reaction mmf implying that the armature reaction mmf is distributed sinusoidally and rotates at synchronous speed with constant magnitude.

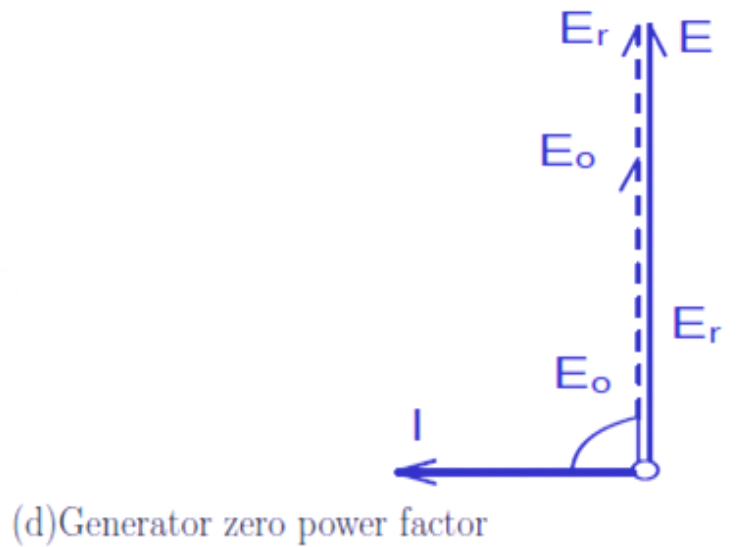
The phasor diagrams for the several conditions contain the phasors of two emfs viz. E_o and E

$$E = E_o + E_r$$

The emf E_r is thus similar to an emf induced in an inductive reactance, so that the effect of armature reaction is exactly the same as if the armature windings had a reactance $x_a = E_r/I_a$. This fictitious reactance x_a can be added to the armature leakage reactance x_l and the combined reactance $(x_a + x_l)$ is known as the synchronous reactance x_s . The behavior of a synchronous machine can be easily predicted from the equivalent circuit developed using this synchronous reactance x_s .



(b) Motor unity power factor



Approximation of the Saturated Synchronous Reactance

Economical size requires the magnetic circuit to be somewhat saturated under normal operating conditions. However, the machine is unsaturated in the **short-circuit test**, and the synchronous reactance based on short-circuits and **open-circuit test** data is only an approximation at best. Nevertheless, there are many studies in which a value based on rated open-circuit voltage and the short circuit current suffices. If V_{oc} is rated voltage, I_{sc} is the required no-load field current, which also produces the armature current I_a on short circuit. The synchronous impedance is, accordingly,

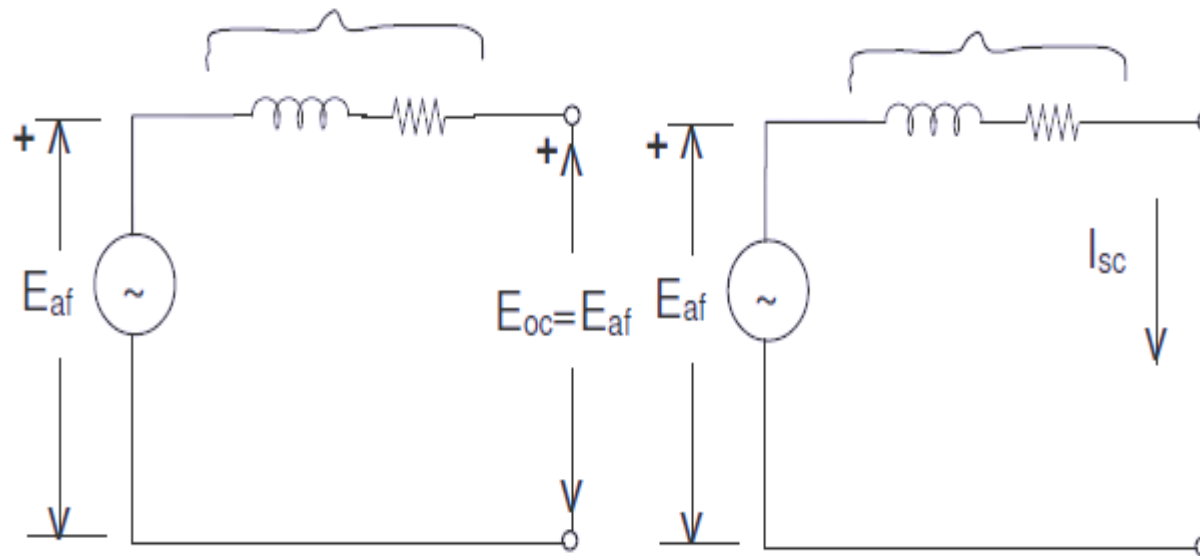
$$Z_{du} = \frac{V_{oc}}{\sqrt{3} I_{sc}}$$

$$X_d = (Z_d^2 - r_a^2)^{\frac{1}{2}} \approx Z_d$$

$$E_{af} = V + Z_d I$$

Open-circuit and Short-circuit Tests

The effect of saturation on the performance of synchronous machines is taken into account by means of the magnetization curve and other data obtained by tests on an existing machine. Only some basic test methods are considered. The unsaturated synchronous impedance and approximate value of the saturated synchronous impedance can be obtained from the open-circuit and short-circuit tests.



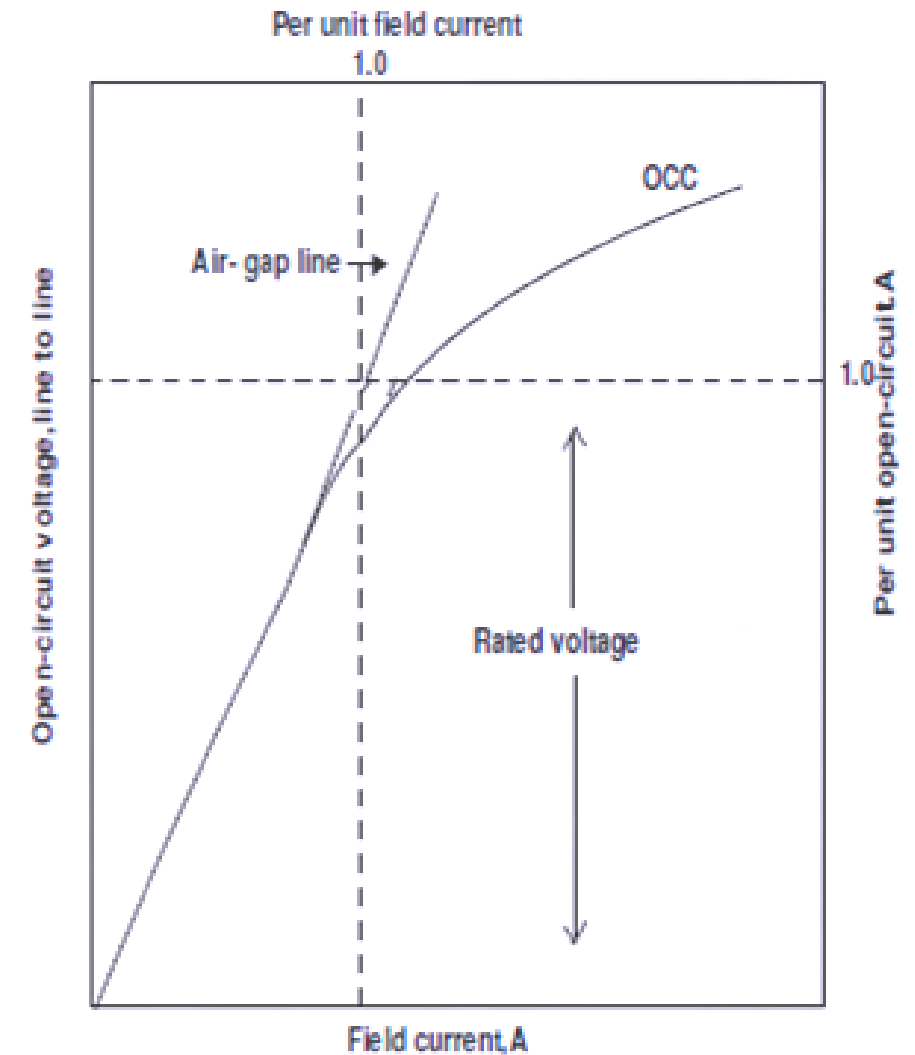
Synchronous generator(a) Open circuit (b) Short circuit

E_{af} is the same in both cases when the impedance $z = E_{af}/I_{sc}$, where E_{af} is the open-circuit volts per phase and I_{sc} is the short-circuit current per phase.

In the case of a constant voltage source having constant impedance, the impedance can be found by dividing the open-circuit terminal voltage by the short circuit current. However, when the impedance is a function of the open-circuit voltage, as it is when the machine is saturated, the open-circuit characteristics or magnetization curve in addition to the short-circuit characteristic is required.

Open-circuit Characteristic

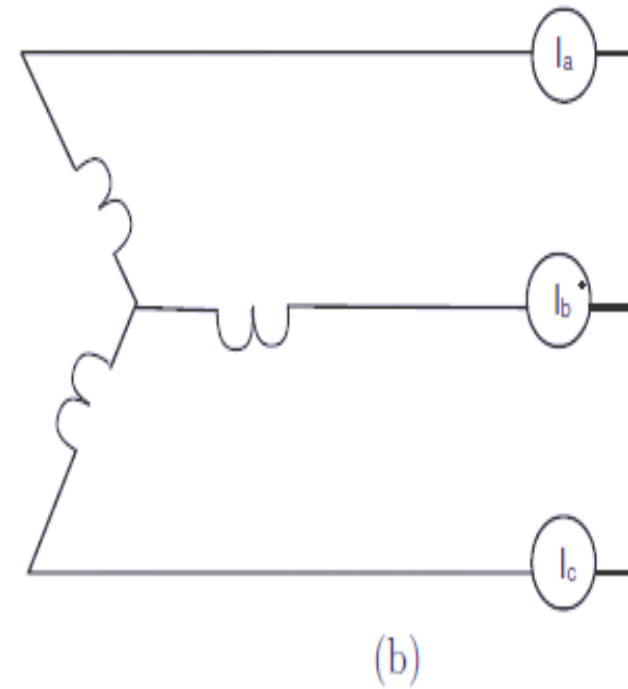
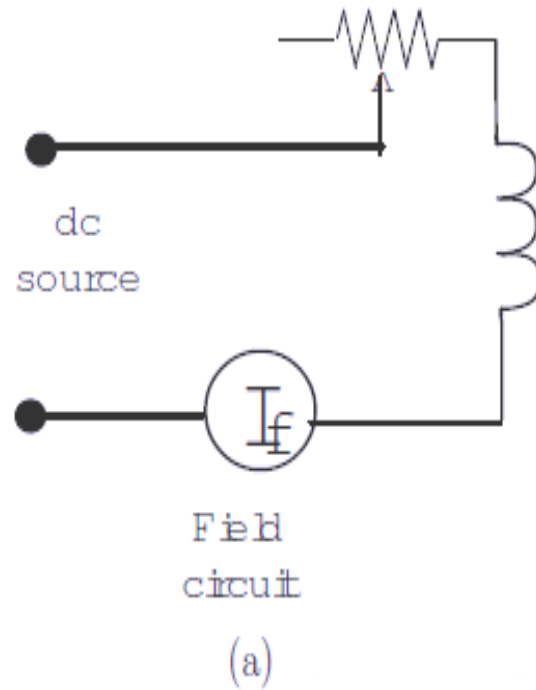
- To obtain the open-circuit characteristic the machine is driven at its rated speed without load. Readings of line-to-line voltage are taken for various values of field current. The voltage except in very low-voltage machines is stepped down by means of instrument potential transformers.
- Two sets of scales are shown; one, line to-line volts versus field current in amperes and the other per-unit open-circuit voltage versus per-unit field current.
- If it were not for the magnetic saturation of the iron, the open-circuit characteristic would be linear as represented by the air-gap line in Fig.
- It is important to note that 1.0 per unit field current corresponds to the value of the field current that would produce rated voltage if there were no saturation.
- On the basis of this convention, the per-unit representation is such as to make the air-gap lines of all synchronous machines identical.



Open circuit characteristic

Short circuit Test

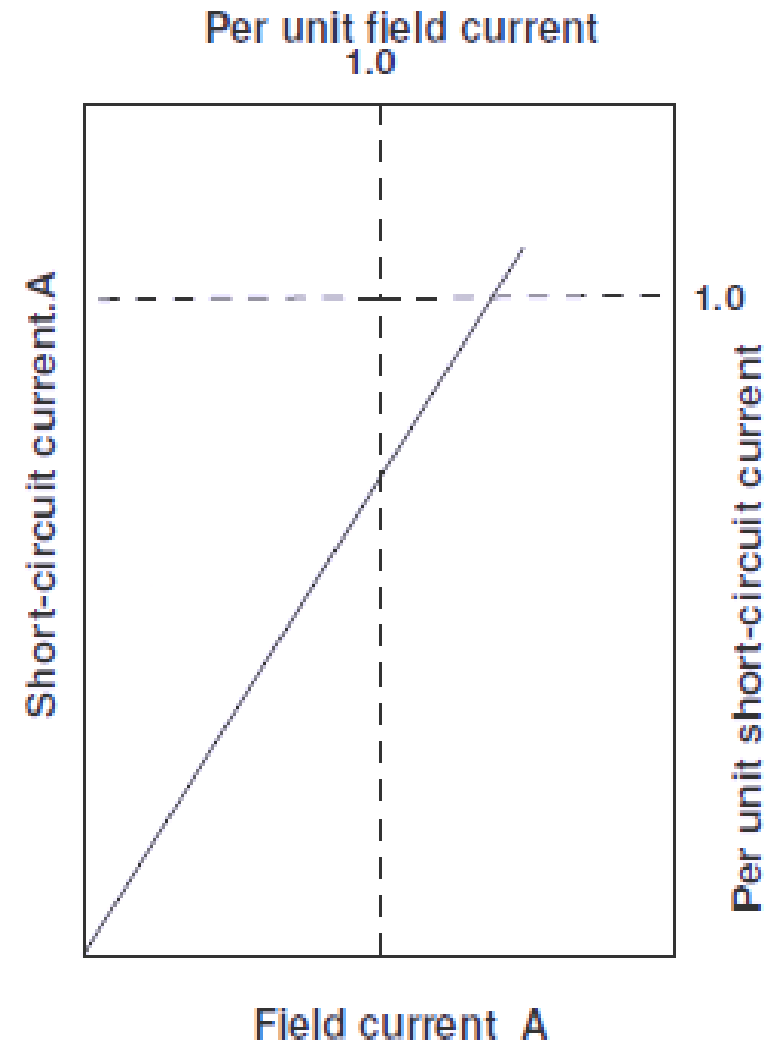
The three terminals of the armature are short -circuited each through a current measuring circuit, which except for small machines is an instrument current transformer with an ammeter in its secondary. A diagram of connections in which the current transformers are omitted is shown in fig.



Connections for short-circuit test

Short-circuit characteristic

In conventional synchronous machines the short-circuit characteristic is practically linear because the iron is unsaturated up to rated armature current and somewhat beyond, because the magnetic axes of the armature and the field practically coincide (if the armature had zero resistance the magnetic axes would be in exact alignment), and the field and armature mmfs oppose each other.



Short-circuit characteristic

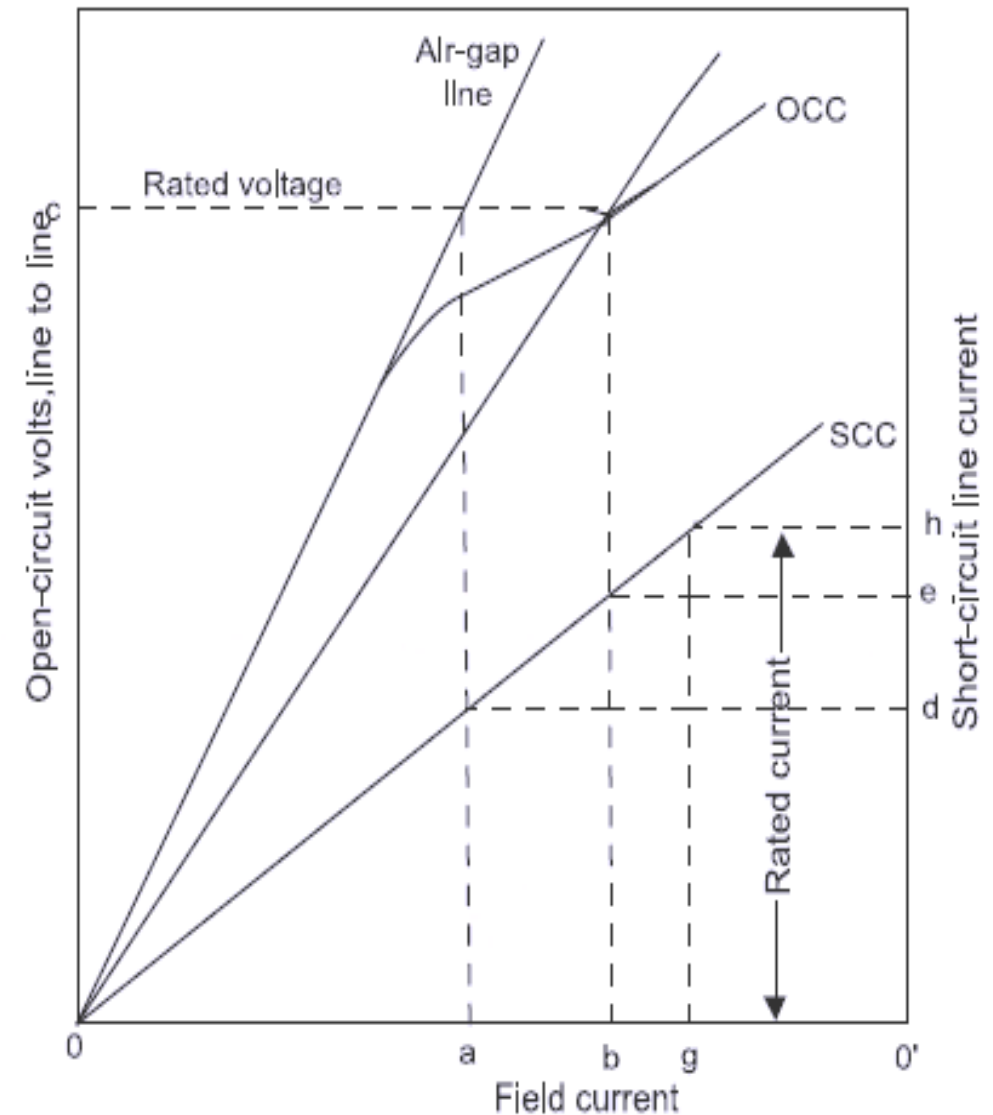
Unsaturated Synchronous Impedance

➤ The open circuit and short-circuit characteristics are represented on the same graph in Fig. The field current Og produces a line-to-line voltage oc on the air-gap line, which would be the open-circuit voltage if there were no saturation. The same value of field current produces the armature current $o'd$ and the unsaturated synchronous reactance is

$$Z_{du} = \frac{oc}{\sqrt{3}o'd} \Omega \text{ phase, for star connected armature}$$

➤ When the open-circuit characteristic, air-gap line, and the short-circuit characteristic are plotted in per-unit, then the per unit value of unsaturated synchronous reactance equals the per-unit voltage on the air-gap line which results from the same value of field current as that which produces rated short-circuit (one-per unit) armature current.

➤ In Figure, this would be the per-unit value on the air gap line corresponding to the field current og .



Open-circuit and short circuit characteristic